Recovering a Circle from Noisy Data

ST5201 Project: Estimation, Denoising and Geometry

Duration: Six weeks, beginning after the 1 October midterm. The project report is due on **12 November**.
Groups: One to three students; all members receive the same project mark.
Programming: Any language is permitted.

The central question

Suppose points lie on a circle, but measurement noise obscures their positions. Can we recover the circle? Can we also recover the original position of each point? This project investigates why these two tasks can lead to different estimators and different conclusions about performance.

The model

Let the unobserved clean points and observed measurements satisfy

$$X_i = c + R(\cos\Theta_i, \sin\Theta_i), \quad Y_i = X_i + \varepsilon_i,$$
$$\Theta_i \sim \text{Uniform}(0, 2\pi), \qquad \varepsilon_i \sim N(0, \sigma^2 I_2).$$

All angles and noise vectors are mutually independent. The circle centre $c \in \mathbb{R}^2$ and radius $R > 0$ are unknown. Treat σ as known in the required experiments. Only the noisy training observations may be used to fit an estimator.

Part A. Estimate the circle (20 marks)

1. Show that $E(Y) = c$ and $E\|Y - c\|^2 = R^2 + 2\sigma^2$. Use these identities to motivate estimates of the centre and radius.

2. With $\widehat{c} = \overline{Y}$, consider
$$\widehat{Q} = \frac{1}{n-1}\sum_{i=1}^{n}\|Y_i - \overline{Y}\|^2 - 2\sigma^2, \qquad \widehat{R} = \sqrt{\max\{\widehat{Q}, 0\}}.$$
Establish the unbiasedness of $\widehat{c}$ and $\widehat{Q}$. Explain the use of $n-1$ and why unbiasedness need not survive truncation or taking a square root. Discuss qualitatively what truncation at zero may do when the sample size is small or the signal-to-noise ratio is low.

3. Explain, using laws of large numbers, why these circle estimates are consistent under the stated model. A short argument is sufficient. Compare the noise-corrected radius with the naive estimate obtained without subtracting $2\sigma^2$.

Connection to the course: moments, expectation, sample variance, bias, consistency and mean squared error. No background in differential geometry is required.

Part B. Implement and understand the methods (30 marks)

Compare the methods below. An **oracle** uses the true circle parameters and is a benchmark unavailable in practice. A **fitted** method estimates its quantities from the noisy training sample. Keep these labels explicit throughout your report.

B1. Projection onto a circle

$$\delta_{\text{proj}}(y; c, R) = c + R\,\frac{y - c}{\|y - c\|}.$$

Implement both oracle projection and projection using $(\widehat{c}, \widehat{R})$ from Part A. Specify a convention at $y = c$ (an event of probability zero under the model); if $\widehat{R} = 0$, the fitted projection returns $\widehat{c}$. Explain what information projection retains and what it discards.

B2. Bayes estimator: posterior mean under the circle model

Under squared Euclidean loss, the Bayes estimator of the clean point is $\delta_{\text{B}}(y) = E(X \mid Y = y)$. Write $x(\theta) = c + R(\cos\theta, \sin\theta)$. With the uniform angular prior, Bayes' rule gives

$$\delta_{\text{B}}(y) = \frac{\int_0^{2\pi} x(\theta)\exp\{-\|y - x(\theta)\|^2/(2\sigma^2)\}\,d\theta}{\int_0^{2\pi} \exp\{-\|y - x(\theta)\|^2/(2\sigma^2)\}\,d\theta}.$$

Special functions are not required. Approximate both integrals on an equally spaced angular grid:

$$\theta_k = 2\pi k/K, \quad x_k = x(\theta_k), \quad k = 0, \ldots, K-1, \qquad a_k(y) = -\frac{\|y - x_k\|^2}{2\sigma^2},$$
$$w_k(y) = \exp\{a_k(y) - \max_{\ell} a_\ell(y)\}, \qquad \delta_{\text{B},K}(y) = \frac{\sum_k w_k(y) x_k}{\sum_k w_k(y)}.$$

Use $K = 512$ and check a representative subset of results with $K = 1024$. Subtracting the largest log-weight avoids numerical underflow. Implement both the oracle rule and the plug-in rule that substitutes $(\widehat{c}, \widehat{R})$. The plug-in rule is not automatically Bayes optimal.

1. Explain how the Gaussian noise model leads to these weights.
2. Use conditional expectation to justify why $E(X \mid Y)$ minimizes expected squared error for the clean point. A brief argument or the standard conditional-expectation identity is sufficient.
3. Plot the output for several observations and noise levels. Does the posterior mean lie on the circle? Explain why its position is compatible with its optimality for squared-error loss.

B3. Manifold fitting (MF) estimator

Why manifold fitting? This circle project is motivated by the broader problem of *manifold fitting*: recovering an unknown low-dimensional shape from noisy observations. A circle is a simple one-dimensional manifold in the plane: one angle determines a clean point, although each observation has two coordinates. Here we know the shape used to generate the data, so we can assess recovery directly. The MF method itself does not impose a circular shape or estimate a centre and radius; it uses local geometric information in the noisy sample. Comparing it with circle-based estimators lets us investigate a central statistical question: *how does the choice between recovering a shape and estimating individual clean points change what counts as a good estimator?*

Optional reading. For a broader, non-technical perspective on geometry, data and trustworthy AI, see my *Straits Times* commentary, *"What a cholera map can teach us about trusting AI"*. This is suggested reading for interest; it is not required for the project.

The MF estimator used here is the two-step weighted contraction map of Yao, Su, Li and Yau (2023), *Manifold Fitting*, arXiv:2304.07680. It uses the noisy training sample $Y_1, \ldots, Y_n$ to map a query point z (here, an independent noisy test observation) to a fitted point. Use $k = 3$ and the fixed radii

$$r_0 = r_1 = \frac{5\sigma}{\log_{10} n}, \qquad r_2 = \frac{10\sigma\sqrt{\log(1/\sigma)}}{\log_{10} n}.$$

Here log is the natural logarithm. These settings are specified for the scales and noise levels in this project ($0 < \sigma < 1$).

Step 1: estimate a contraction direction. Define

$$\widetilde{\alpha}_i(z) = \begin{cases} \left(1 - \dfrac{\|Y_i - z\|^2}{r_0^2}\right)^k, & \|Y_i - z\| \le r_0, \\ 0, & \text{otherwise,} \end{cases} \qquad F(z) = \frac{\sum_i \widetilde{\alpha}_i(z) Y_i}{\sum_j \widetilde{\alpha}_j(z)},$$

and set $e(z) = \{F(z) - z\}/\|F(z) - z\|$ and $U(z) = e(z)e(z)^\top$.

Step 2: weighted cylindrical contraction. Write $u_i = U(z)(Y_i - z)$ and $v_i = (I_2 - U(z))(Y_i - z)$. The cylinder has half-length r_2 along $e(z)$ and radius r_1 orthogonal to it:

$$w_u(u) = \begin{cases} 1, & \|u\| \le r_2/2, \\ \left[1 - \left(\dfrac{2\|u\| - r_2}{r_2}\right)^2\right]^k, & r_2/2 < \|u\| < r_2, \\ 0, & \text{otherwise,} \end{cases} \qquad w_v(v) = \begin{cases} \left(1 - \dfrac{\|v\|^2}{r_1^2}\right)^k, & \|v\| \le r_1, \\ 0, & \text{otherwise.} \end{cases}$$

With $\widetilde{\beta}_i(z) = w_u(u_i) w_v(v_i)$, the MF output is

$$\widehat{\delta}_{\mathrm{MF}}(z) = G(z) = \frac{\sum_i \widetilde{\beta}_i(z) Y_i}{\sum_j \widetilde{\beta}_j(z)}.$$

Degenerate cases. With small samples a neighbourhood can be empty. Use these conventions rather than dropping a query: if $\sum_j \widetilde{\alpha}_j(z) = 0$, return z; if $\|F(z) - z\|$ is numerically zero, return $F(z)$; if $\sum_j \widetilde{\beta}_j(z) = 0$, return $F(z)$. Report the proportion of queries for which a convention is used.

A reference implementation of this MF map is supplied. You may use it directly. In your report, explain in words why the first neighbourhood estimates a direction and why the second is a cylinder, document the settings and fallback rate, and evaluate the map only on independent queries. You are not required to reimplement MF, derive it, or prove its convergence.

Part C. Common simulation study (25 marks)

Use $c = (0, 0)$, $R = 1$, sample sizes $n \in \{100, 500, 2000\}$ and noise levels $\sigma \in \{0.05, 0.15, 0.30\}$. Fit the estimators without supplying the true c or R, except to the explicitly labelled oracle benchmarks.

For reproducibility, set the random seed to **5201** once at the start of your simulation script. Do not reset the seed separately for different methods.

For each of the nine settings, use **50 independent repetitions**. In each repetition, generate a fresh noisy training sample and an independent test sample of $m = 500$ clean/noisy pairs. Apply all methods to the same test sample within a repetition. More repetitions are optional, not a source of extra marks.

Compare: (i) the raw noisy observations (the rule $z = y$), (ii) oracle projection, (iii) fitted projection, (iv) the oracle posterior mean, (v) the plug-in posterior mean, and (vi) MF.

C1. Parameter recovery and pointwise denoising

For the circle estimator, report centre MSE, radius bias and radius MSE across repetitions, and how often $\widehat{Q}$ is negative. MF does not estimate (c, R), so no centre or radius error is reported for it.

For each rule, let z_j be its output for test observation Y_j^{test}. Report

$$\text{Denoising MSE} = \frac{1}{m}\sum_{j=1}^{m}\|z_j - X_j^{\text{test}}\|^2.$$

C2. Geometric recovery

Report both the distance of the outputs to the circle and their coverage of it. For a fixed grid of $L = 720$ equally spaced true-circle points q_ℓ, compute

$$\text{Radial error} = \frac{1}{m}\sum_{j=1}^{m}\big(\|z_j - c\| - R\big)^2, \qquad \text{Coverage error} = \frac{1}{L}\sum_{\ell=1}^{L}\min_{1\le j\le m}\|q_\ell - z_j\|^2.$$

Coverage error contains a spacing effect from using only m output points, so keep m and L fixed and report the coverage error of the clean test points X_j^{test} as a reference baseline, not a strict lower bound. Explain why a small radial error alone does not establish recovery of the entire circle.

C3. Evidence and interpretation

Report mean performance and Monte Carlo standard errors calculated from the 50 repetition-level results; do not treat all test points from all repetitions as independent experiments. Include representative equal-axis scatterplots and plots showing how error changes with n and σ.

Clean points, true circle parameters and test losses must not be used to tune a fitted method; use the MF settings of Part B3. Do not select only favourable repetitions.

Discuss whether method rankings change with the loss. Bayes optimality concerns expected pointwise squared error under the correct model and true parameters; it does not imply optimal geometric recovery. Small empirical reversals may reflect simulation or quadrature error. There is no requirement that MF outperform any comparator.

Part D. An open investigation (25 marks)

Choose **one** of the following questions, or formulate a closely related one. State a conjecture and design experiments that could challenge it. A convincing negative finding can receive full credit; a new method or theorem is not expected.

Option 1. What can more training data fix?

Keep σ fixed and extend the range of n. Investigate which errors decrease, which appear to level off, and why. Distinguish error from estimating the circle, the behaviour of a fixed denoising rule, and the irreducible uncertainty about a clean point given one noisy observation. Compare the fitted methods with their oracle benchmarks, keeping the test sample size fixed. A useful question is whether the plug-in posterior mean approaches oracle performance while its output still differs geometrically from the true circle. Do not infer a nonzero limiting error from a short flat segment alone.

Option 2. What if points are sampled unevenly?

Replace uniform angular sampling by the density

$$f_a(\theta) = \frac{1 + a\cos\theta}{2\pi}, \qquad 0 \le \theta < 2\pi, \quad 0 \le a < 1,$$

for example with $a \in \{0, 0.5, 0.9\}$. Generate angles by rejection sampling from the uniform distribution, accepting θ with probability $(1 + a\cos\theta)/(1 + a)$. Explain why the observed-data mean need not equal the circle centre.

Evaluate how this change affects moment-based circle fitting, MF and the uniform-prior posterior-mean rule. The latter is now a misspecified rule, not the Bayes optimum for this model; for a correctly specified oracle comparison, multiply each quadrature weight by $1 + a\cos\theta_k$. Examine both n and a, and discuss the effect of sparse angular regions on coverage.

Option 3. What if the shape is only approximately a circle?

Generate clean points from an ellipse,

$$X = c + (\cos\Theta,\ b\sin\Theta), \qquad \Theta \sim \text{Uniform}(0, 2\pi), \quad b \in \{1, 0.9, 0.75\}.$$

Keep the fitted circle-based methods and MF, and investigate the cost of imposing a circular shape. Vary b and n at a fixed noise level. Uniform angle does not mean uniform arc length; retain the stated sampling mechanism.

Evaluate geometry using distances to a dense ellipse grid in both directions; the circle radial formula no longer applies. A posterior mean based on the circle model is misspecified. A true-ellipse posterior mean, computed by replacing the circle grid of Part B2 with the true ellipse grid, may be included as an oracle benchmark.

Requirements for the open investigation

1. State the question, target and conjecture before the final experiment.
2. Vary at least two relevant factors using a small, informative design.
3. Report uncertainty, explain one limitation, and distinguish observed evidence from claims about all distributions or arbitrarily large samples.
4. End with a concise answer to your question. Explain what you learned that was not already apparent from the common study.

Final reflection

Conclude your report with at most 200 words answering: **"If someone asks you to recover a circle from noisy data, what must you clarify before deciding which estimator is best?"** Support your answer with the distinction between parameter recovery, pointwise denoising and geometric recovery.

Submission

Report: One PDF of at most 10 pages, including figures and tables but excluding the title page and references. An optional appendix of at most 3 pages may contain supplementary details; the main argument must be understandable without it. Before submission, email zhigang.yao@nus.edu.sg with your name and the names of all other group members (if any). A folder will then be created for your submission on Canvas.

Code and results: Submit code, a README and the numerical summaries underlying the main figures in one archive. State software versions and instructions for reproducing the results. Separate data generation, fitting and evaluation clearly. Do not submit only screenshots of program output.

Group contribution statement: Briefly describe each member's contribution. Groups of one, two or three students have the same requirements.

AI use: AI assistance with coding, debugging and editing is permitted if disclosed. Explain how substantial assistance was checked. Each member is responsible for understanding the methods, verifying the results and acknowledging sources.

Suggested six-week schedule

Week	Milestone
1	Understand the model; complete Part A; implement and check data generation.
2	Implement oracle and fitted circle projection and posterior means.
3	Integrate the supplied MF map; check evaluation on independent queries and the degenerate-case reporting.
4	Complete the common simulation study; formulate the open question.
5	Carry out the open investigation and final runs.
6	Interpret the findings, check reproducibility and prepare the report.

These milestones are guidance, not separate graded submissions.

Clarity and reproducibility are assessed within the relevant components. More complicated algorithms, more plots, or favourable results for a particular method do not automatically earn higher marks.